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Creating 3D printed magnetic devices with ferrofluids and liquid metals

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ABSTRACT

Combining electrical and magnetic materials in the same part has been a challenge in 3D printing due to difficulties co-printing complex materials in many additive manufacturing processes. Past 3D printed inductors and other similar magnetic devices have therefore either lacked the magnetic materials necessary for improved performance, or required sintering at high temperatures for extended periods, beyond the capability of most 3D printable polymers. In this work, we demonstrate a room temperature process for incorporating conductive and magnetic materials into the same 3D printed device. A multi-stage fabrication process based on 3D printing followed by fill with magnetic and conductive fluids is proposed. Multi-layer microfluidic channels for magnetic passives are first printed in a stereolithography process. The microfluidic systems are then filled with room temperature liquid metal, a gallium alloy liquid at room temperature, and ferrofluid to create inductors, transformers and wireless power coils. Through the addition of ferrofluid as a magnetic material, increases in inductance density by nearly a factor of three were demonstrated, in addition to coupling improvements for transformer and wireless power coils compared to the devices before magnetic fill.

1. Introduction

Incorporation of electronics is one of the next frontiers in 3D printing, allowing electrical systems to be direct printed into complex shapes [1]. Building parts layer by layer gives rapid customization, enables complex structures difficult or impossible by other means, and reduces waste in manufacturing [2]. Conductors can be incorporated into 3D printed parts through a variety of approaches ranging from deposited conductive inks to integrated metal sheeting or wiring [3]. An important application is for passive devices and fundamental circuit elements such as capacitors and inductors. Using additive manufacturing allows designers to create unusual geometries difficult or impossible to achieve by other means for improved electrical performance [4]. Devices can be directly incorporated into the body of a complex structure [5], giving improved use of mass and volume in cases where size or form factor are critical. 3D printing also allows for electrical devices highly customized for a specific end user application [6].

Inductors and transformers, components of power and communication systems, are crucial to most electronics and tend to scale poorly with size [7], resulting in a large volume to obtain better performance. There is therefore particular value to 3D printing an inductor, both to allow for improved device geometry as well as to better integrate the inductor into the casing or physical structure to improve weight and reduce size. The biggest challenge, and the focus of most of the research in the area, has been on the quality of electrical conductors available, and there have been great strides made in the past few years. An inductor is fundamentally a coil of wire, and in theory any 3D printed conductor can be used to make one. 3D printed inductors have been demonstrated through conductive thermoplastic filament [8], silver paste [5,9], metal casting [4], liquid gallium alloys [10,11], and electro- [12] and electroless [13,14] metal plating. It is now possible to fabricate a 3D printed inductor with trace conductivity approaching or even equal to those made through traditional means.

One area where 3D printed inductor technology has lagged, however, is in incorporation of magnetic materials. High permeability materials are routinely used in traditional inductor design, resulting in improved coupling, energy storage and inductance density [15]. Despite its promise, use in a 3D printed inductor remains rare, and there are only a handful of examples available in the literature. One possibility is to physically insert a separately manufactured magnetic core into the final 3D printed inductor [5], which is effective but only possible for limited geometries. There have also been several examples of printing only the magnetic core followed by manually wrapping with wire to form an inductor [16,17].

To our knowledge the only demonstrated example of a fully integrated 3D printed magnetic core inductor has been recent work from Virginia Tech on using magnetic and conductive pastes printed with extrusion printing [9]. The two pastes are first deposited by a dual head

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extrusion 3D printer, followed by heating to very high temperature (250 °C) for an extended period (30 min at 250 °C, plus four hours ramping up and down temperature) for curing. Although the performance was impressive, achieving magnetic relative permeability of 23, the need to heat to temperatures higher than many printed polymers can survive means that this approach is not well suited to most additive manufacturing technologies. There is still a strong need for a better approach to incorporating magnetic materials into 3D printed inductors and transformers.

In this work, we demonstrate a room temperature alternative: a fluidic 3D printed magnetic core inductor. Liquid metals are combined with a magnetic core based on ferrofluid, creating complete magnetic core 3D printed inductors in a room temperature process. Using this technique, we demonstrate a variety of common inductor topologies, including basic magnetic core inductor designs (a toroid and solenoid), a magnetic transformer, as well as the use of ferrofluid for improved coupling for wireless power transfer to a 3D printed inductor.

2. Methods and materials

2.1. Fabrication

In this work, a low cost commercial stereolithography 3D printer (Form 2, Formlabs) was chosen as a representative and widely available 3D printer. In stereolithography, a laser is scanned across a bath of photoresin to selectively solidify the resin and build up polymer parts [19]. Microfluidic channels with widths as small as 300 μm have been demonstrated using a standard stereolithography 3D printing process [20]. Switching to a higher resolution micro-stereolithography process [21] would also allow the minimum possible channel width to be reduced further. The process has also previously been demonstrated for making electrical devices through injection of liquid metal, including Helmholtz coils [11] and foldable antennas [22], and is therefore well suited for creating the magnetic core inductors and transformers here.

Fig. 1 (a) shows a transformer after printing. In stereolithography, liquid resin becomes trapped in a fully enclosed channel, and the uncured resin must then be drained from the part [20] (Fig. 1 (b)), followed by fill with liquid metal and ferrofluid (Fig. 1 (c)). SubMiniature A (SMA) connectors, a common form of RF electrical connector [24], are then added into patterned sockets in the 3D printed part to allow for electrical characterization (Fig. 1 (d)).

2.2. Ferrofluid

With filling a channel with a conductor such as liquid metal or conductive paste a well-established method for creating a 3D printed inductor, using similar channels with a magnetic fluid is a promising avenue to provide a magnetic core inductor without a significant change in the fabrication process. Several groups have examined ferrofluids for magnetic cores of inductors, including stretchable electronics [25], microelectromechanical systems (MEMS) [26], and macroscale transformers [27]. Ferrofluids are magnetic fluids consisting of nanoscale magnetic particles in a carrier fluid [28]. By keeping particle sizes very small (less than 15 nm), and combining with a suitable surfactant to prevent agglomeration, Brownian motion is sufficient to prevent settling and provide a uniform distribution [29].

A number of commercial formulations are available for ferrofluid, and two different ones were investigated here, both from the supplier FerroTec, with properties shown in Table 1. EFH1 is marketed as a lower grade educational ferrofluid, while EMG900 is a higher permeability ferrofluid targeting the audio market. The same two ferrofluids were investigated in our previous work on creating a stretchable ferrofluid inductor [25], allowing direct comparison. In general, a higher core permeability of EMG900 is expected to result in a higher inductance than EFH1, but at a higher material cost; EFH1 is roughly a factor of 30 lower cost per mL than EMG900, and would be preferable in circumstances where higher cost is impractical but a more moderate increase in inductance would still be beneficial.

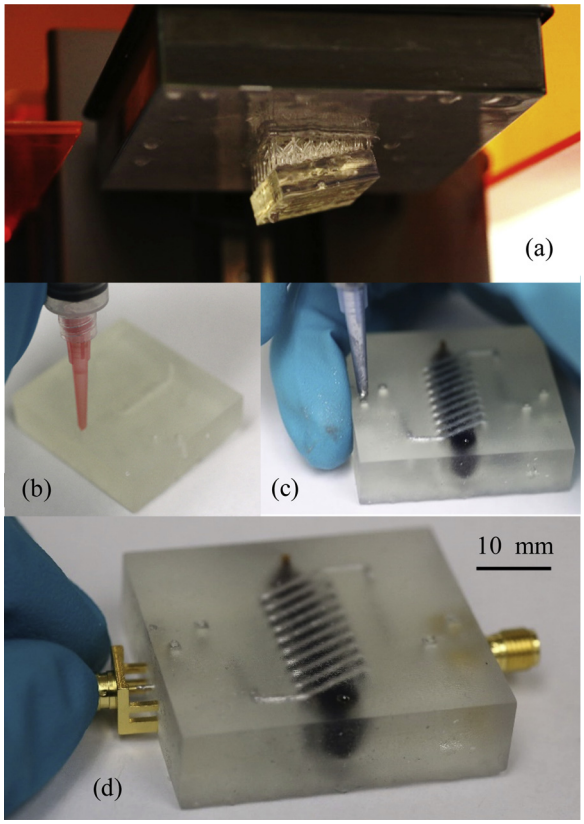


Fig. 1. Ferrofluid-cored 3D printed inductor fabrication process: (a) after printing, (b) draining of uncured resin, (c) addition of liquid metal and ferrofluid and (d) addition of SMA connectors.

Table 1
Properties of ferrofluids used (from manufacturer datasheet) [30].

Ferrofluid	Carrier Liquid	Particle Material	Relative Permeability	Saturation Magnetization
EFH1	Hydrocarbon	Magnetite	3.6	44 mT
EMG900	Hydrocarbon	Magnetite	19.6	99 mT

2.3. Liquid metal

Creating an inductor through 3D printed fluidic channels requires a liquid conductor, and both of the two choices demonstrated by the community (liquid metals and conductive pastes) have their advantages. Though silver paste can reach very high conductivities, the best performance is only possible with post printing processing through high temperature sintering as in ref. [9], hotter than many 3D printable polymers. Liquid metals were chosen here in light of our goal of a low temperature process. The room temperature liquid metal gallium alloys are widely used as a non-toxic alternative to mercury in applications such as reconfigurable or printable antennas [31] and stretchable electronics [32]. Galinstan, the alloy here, has resistivity of 2.9×10^{-7} Ω·m and a melting point of −19 °C [33].

3. Results and discussion

3.1. Liquid metal inductors

In ref. [11], a liquid metal solenoid was printed using a similar

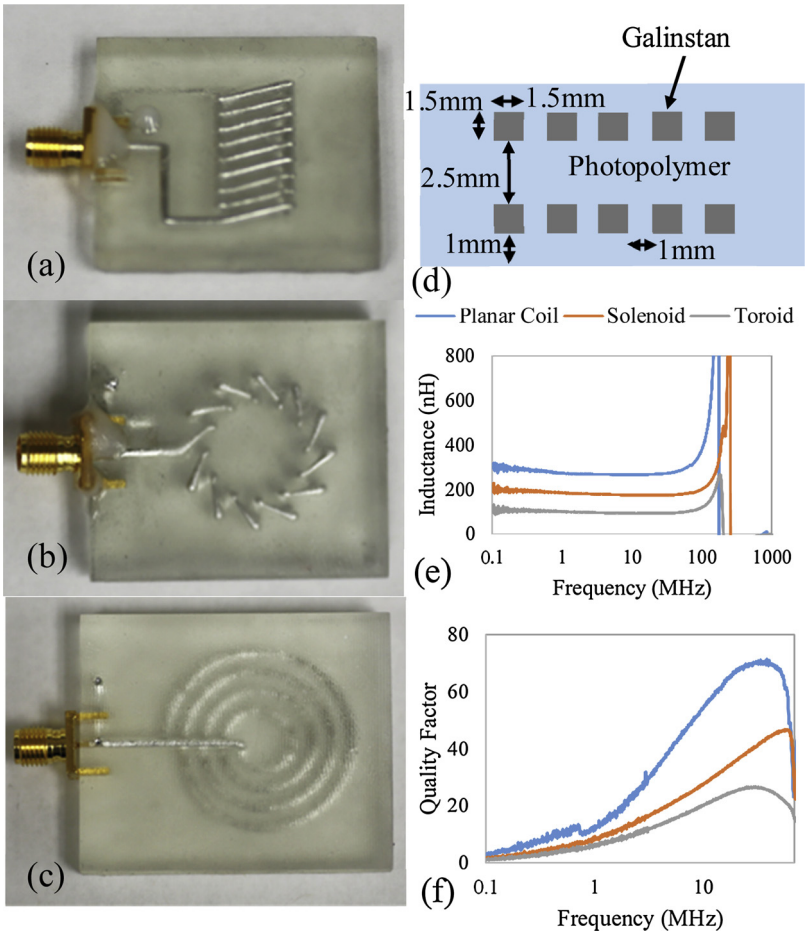


Fig. 2. 3D printed liquid metal (a) 9-turn solenoid, 12-turn toroid and (c) 4 turn planar coil, (d) cross section with dimensions and (e) measured inductance and (f) quality factor.

approach as part of a Helmholtz coil for electromagnetic field generation, but the electrical behavior of the inductor was not characterized. To understand the behavior, three non-magnetic core inductor designs were fabricated and characterized to extract the inductor performance, a 9-turn solenoid (Fig. 2 (a)), a 12-turn toroid (Fig. 2 (b)) and a 4-turn planar coil (Fig. 2 (c)). A representative cross section is shown in Fig. 2 (d). The devices consisted of two conductive layers with vertical vias, with the conductive layers 2.5 mm apart, and 1 mm of polymer above and below the liquid metal channels. While smaller channels were printable, 1.5 mm on a side cross section was found most consistent for the longer channels necessary to make a full inductor on this 3D printer, and this channel size was used for the liquid metal traces in the remainder of this work. This is a limitation of the cheaper consumer grade printer used here, and smaller channels have been demonstrated elsewhere with more powerful stereolithography tools [20].

The inductors were characterized using a Keysight E5061B network analyser, with the inductance and quality factor measurements for all three designs shown in Fig. 2 (e) and (f) respectively. The quality factor is a standard measure of how efficiently an inductor stores magnetic energy relative to its resistive losses, and is defined as the ratio of the inductor's reactance (the imaginary component of its impedance) to its resistance (the real component of its impedance) at a given frequency [34]. Therefore, a high quality factor means more energy can be stored relative to the inductor's losses, and is generally preferred in inductor design. At high frequencies, the inter-winding capacitance begins to resonate with the device inductance, resulting in a maximum value slightly below self-resonance known as the peak quality factor (the maximum value of the quality factor curve, for instance in Fig. 2(f)). The inductor peak quality factors were 46.7 ± 0.6, 26.6 ± 0.1, and 71.3 ± 0.3 for the solenoid, toroid and planar coil respectively, occurring at 60.7 MHz, 28.6 MHz and 38.9 MHz.

3.2. Ferrofluid-cored inductors

Following characterization of the liquid metal printed inductors, several inductor designs were fabricated with an embedded fluidic chamber to incorporate a ferrofluid core. First, a liquid metal solenoid was designed with 9 turns and a core consisting of a 7.5 mm wide, 6 mm tall cavity for ferrofluid fill (Fig. 3 (a), (b), and (c)). Due to the need for fluidic isolation between the liquid metal and the ferrofluid, walls 0.75 mm wide were defined between the ferrofluid cavity and the liquid metal traces, giving a total core area of 67.5 mm^2, of which 67% of the area is filled with the magnetic fluid. These dimensions are partially defined by the printer limitations; a higher-grade stereolithography printer could achieve better print resolution. With better print resolution, a narrower wall would be possible, allowing the core area to be more completely filled with magnetic material. This would increase the inductance of the inductor further.

At lower frequencies (below 500 kHz), the unfilled inductor was found to have an inductance of 365 ± 10 nH, which increased to 495 ± 8 nH and 950 ± 12 nH for the EFH1 and EMG900 ferrofluids respectively (Fig. 3 (d)) after a calculated fixed inductance for the input traces was subtracted. As expected, adding ferrofluid increases the inductance, and the higher permeability of EMG900 increases the inductance by significantly more than the lower permeability ferrofluid. The quality factor of the inductor (Fig. 3 (e)) was found to be similar at lower frequencies for all three solenoids, and at higher frequencies above 3.4 MHz the unfilled inductor has a superior quality factor while

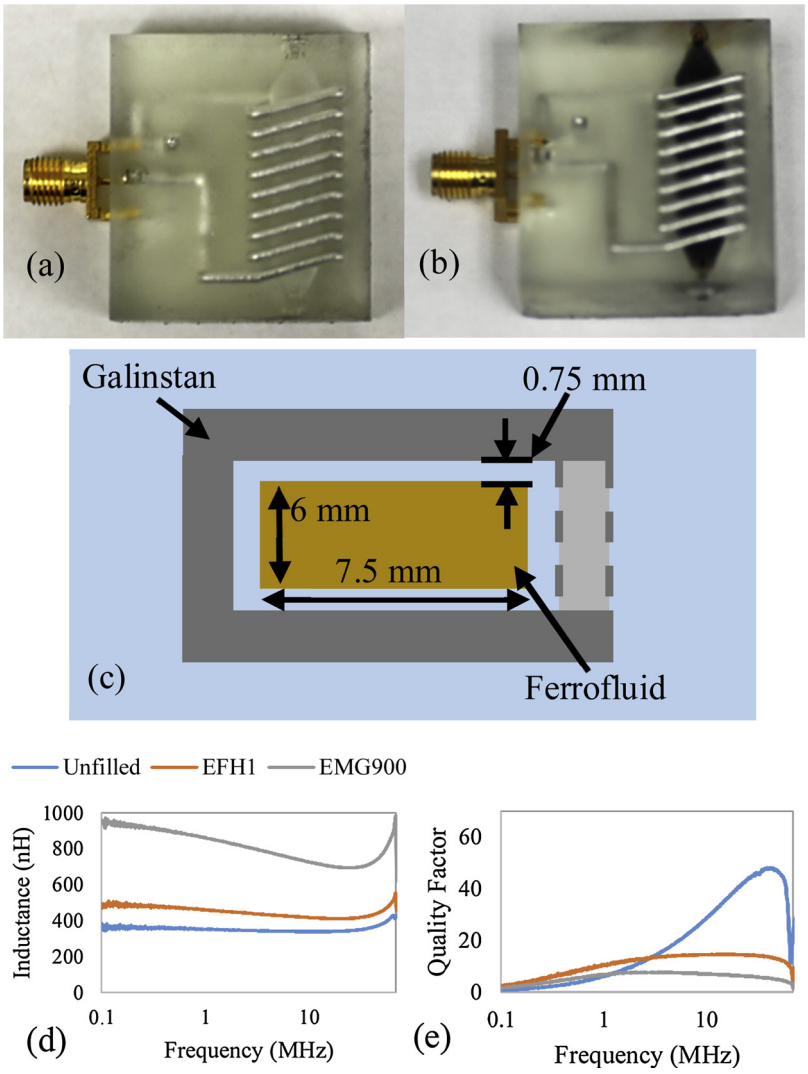


Fig. 3. Ferrofluid-cored solenoid (a) before and (b) after fill with ferrofluid, (c) schematic cross section of solenoid, and measured (d) inductance and (e) quality factor.

the ferrofluid inductors both begin to become more lossy. This result means that one aspect of inductor performance, the inductance density (inductance per volume) has improved, allowing more energy storage in the device. The quality factor (how efficiently the inductor stores energy) remains unchanged (neither better nor worse than before ferrofluid fill) for lower frequencies. The drop in quality factor at higher frequency is characteristic of most magnetic inductor core materials [15], resulting in air-core inductors being more common for high frequency applications, and has also been seen previously for past work with ferrofluids [26], including for the specific ferrofluids here in our previous work [25].

A more complex ferrofluid-cored toroidal inductor was also printed, with similar results (Fig. 4). The toroid consists of 12 liquid metal turns around a core consisting of ferrofluid in 52% of its cross sectional area. The inductance increased from 260 ± 14 nH before fill to 335 ± 11 nH and 520 ± 8 nH for the EFH1 and EMG900 ferrofluids respectively, again after subtracting a fixed estimate of the input trace inductance. Similar behavior was found for both the level of inductance increase and for the quality factor as for the solenoid for both the two ferrofluids as shown.

For comparison, the current state of the art in 3D printed magnetic core inductors is the cured magnetic paste work from Virginia Tech in [9], where after sintering at 250 °C a relative permeability of 23 was demonstrated. The relative permeability of the EMG 900 ferrofluid in this work is similar (19.6). The requirement for a layer of non-magnetic polymer in between the conductive and magnetic fluids results in a lower increase in inductance than would be possible if the traces were completely surrounded by magnetic material. As filled, the inductance density increases by a factor of nearly three compared with before fill, a useful improvement though not as large as in the Virginia Tech work. This result is however obtained in a room temperature process, which enables use in a broader range of 3D printing processes. A larger improvement would also be possible with use of higher resolution printing process such as microstereolithography to reduce the necessary non-magnetic material.

3.3. 3D printed transformer

Transformers are another widely used magnetic component, used for voltage conversion and isolation in power electronics. A transformer consists of two coupled inductors, and magnetic cores are commonly used in traditional transformer design as a method to improve coupling between the two individual inductors. Using the same fabrication process, 3D printing of a magnetic core transformer was also demonstrated (Fig. 5 (a)). The transformer consists of two interwound solenoids, each with 5 turns, with total core area 67.5 mm^2, of which 40% of the area is filled with magnetic fluid.

The 3D printed inductor was characterized by extracting the two

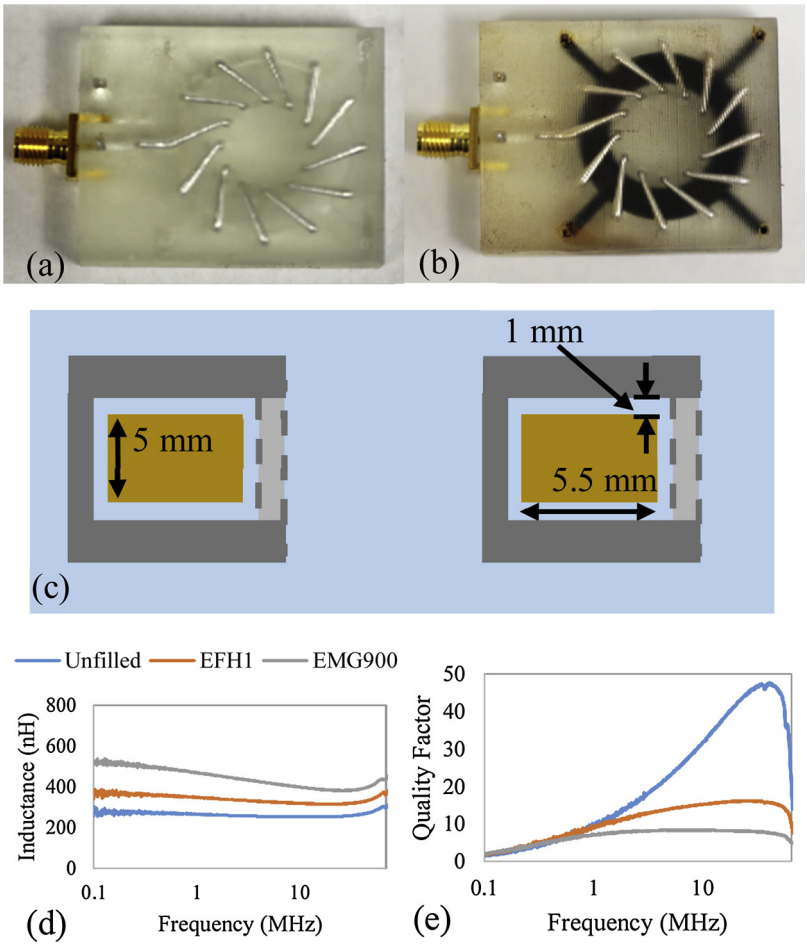


Fig. 4. Ferrofluid-cored toroid (a) before and (b) after fill with ferrofluid, (c) schematic cross section diagram, and measured (d) inductance and (e) quality factor.

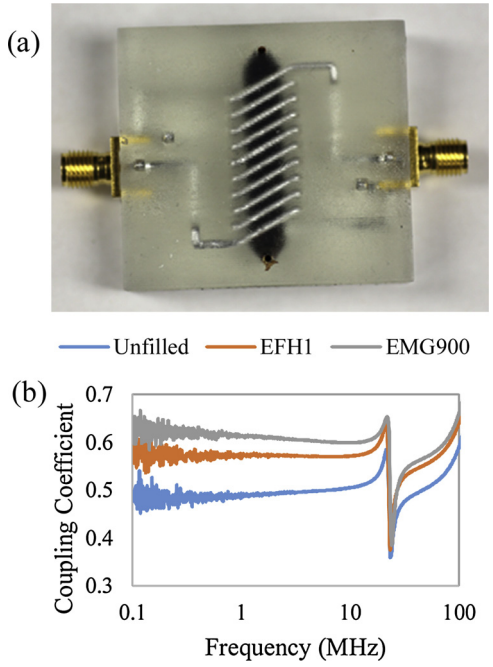


Fig. 5. Ferrofluid-cored transformer and (b) measured coupling coefficient.

port s-parameters [35] using the network analyser and extracting the self and mutual inductances of the two individual inductors. One of the primary figures of merit of a transformer is the coupling coefficient [36]:

$$k = \frac{M}{\sqrt{L_1 L_2}} \tag{1}$$

where L_1 and L_2 are the self-inductance of the two coils, and M is the mutual inductance. The self-inductance can be defined as the inductance of a coil in isolation, while mutual inductance is defined by the magnitude of voltage generated in a second coil through a current in the first. Before fill, the low frequency self-inductance was measured to be 157 ± 9 nH and 163 ± 9 nH for coils one and two respectively, while the mutual inductance was 78 ± 1 nH. With EFH1, the self-inductance was 190 ± 9 nH and 200 ± 13 nH for coils one and two, with mutual inductance 113 ± 1 nH. For the EMG900 fill, the self-inductance was 237 ± 10 nH and 240 ± 8 nH, and the mutual inductance was 150 ± 1 nH. The coupling coefficient is a measure of how strongly coupled the two coils are, and can vary between +1 (perfect positive coupling) and -1 (perfect negative coupling), with 0 indicating no coupling between the coils. The coupling coefficient was extracted for the transformer before and after fill with ferrofluid (Fig. 5 (b)). The low frequency coupling coefficient was found to increase from 0.47 ± 0.02 before fill to 0.57 ± 0.02 and 0.62 ± 0.02 with the EFH1 and EMG900 ferrofluids respectively.

For comparison, a large scale industrial transformer is capable of reaching coupling coefficient values on the order of 0.99 or higher [37], but a smaller transformer with fewer turns as would be used in a small scale power converter, like the one here, will have lower coupling on the order of 0.1 to 0.7 [38,39]. Past work on using ferrofluid in a MEMS transformer, for instance, demonstrated an improvement in coupling coefficient from 0.1 before ferrofluid fill to 0.12 after [26]. The increase

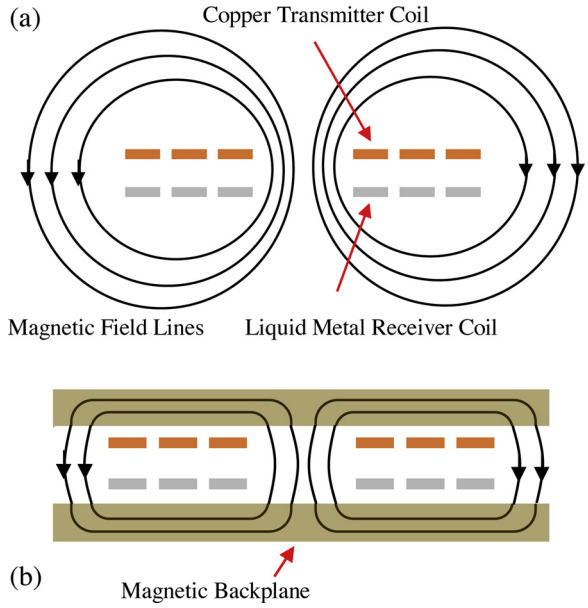


Fig. 6. Inductive wireless power system (a) without and (b) with magnetic backplanes.

with ferrofluid achieved here (0.47 to 0.62) is sizeable in this context, given that 1 is the maximum possible, and well above measurement uncertainty.

3.4. Wireless power backplane

In wireless power, magnetic materials are commonly used to serve as a magnetic backplane, a layer of magnetic material behind a wireless power coil. Adding a high permeability layer serves to guide magnetic fields and prevent radiative fields, as shown in Fig. 6. As a result, magnetic backplanes result in increased coupling between coils and minimize undesired energy losses through leakage fields [40,41].

Based on this principle, a 3D printed inductor was designed to incorporate a ferrofluid magnetic backplane for wireless power. The inductor consists of a 3 turn planar coil with outer diameter 30 mm (Fig. 7 (a)) below a similarly sized pocket which is subsequently filled with ferrofluid to create the magnetic backplane (Fig. 7 (b)). A spacing of 1.5 mm is defined between the inductor and the backplane, with a cut-out region of the ferrofluid pocket used to allow an overpass for the liquid metal inductor.

Following fabrication, the coupling was measured between the 3D printed magnetic backplane inductor and a commercial wireless power coil with a ferrite backplane (Wurth Electronics 760308111, nominal inductance 6.3 μH) (Fig. 7 (c)). The coupling coefficient between the coils was 0.51 ± 0.01 at low frequencies before increasing to 0.53 ± 0.01 and 0.56 ± 0.01 for the EFH1 and EMG900 ferrofluids respectively. The magnitude of increase of the coupling coefficient is

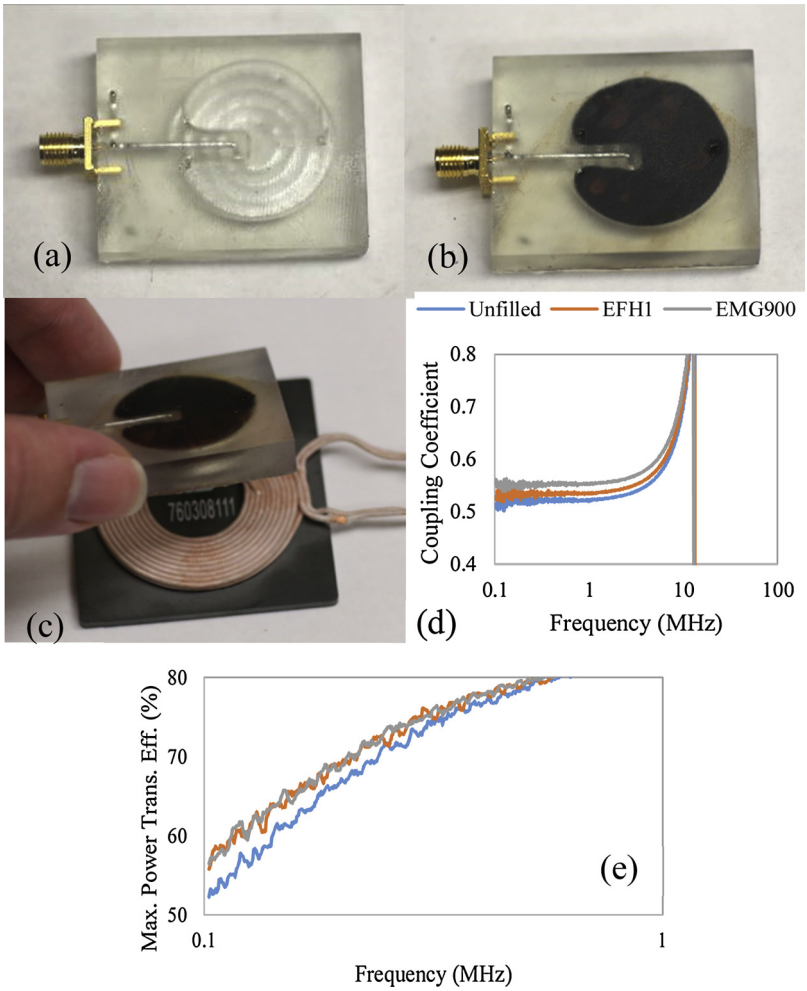


Fig. 7. Planar coil with magnetic backplane (a) before and (b) after fill with ferrofluid, (c) during wireless power testing and (d) measured coupling coefficient and (e) maximum power transfer efficiency.

consistent with our past work on using a different relatively low permeability material (a magnetic polymer composite) in a similar manner to make a stretchable inductor with a magnetic backplane [41]. In our prior work, we found a coupling coefficient increase of 0.06, similar to the level here, by adding a backplane directly against the stretchable inductor with relative permeability of roughly 3. A higher permeability rigid ferrite plane was found to increase the coupling by 0.13 in a similar geometry.

For wireless power, the coupling coefficient is related to but is not equal to the efficiency of the system. For an inductive wireless power systems the maximum power transfer efficiency, the efficiency to a matched load for the two coils in resonance, is [42]:

$$\eta_{\max} = \frac{k^2 Q_1 Q_2}{(kQ_1 + k^2 + 1)(kQ_2 + 1)} \quad (2)$$

where Q_1 and Q_2 are the quality factors of the transmitter and receiver coils respectively.Since the quality factor of a ferrofluid loaded inductor drops off at higher frequencies, the power transfer efficiency was improved only at lower frequencies (below approximately 500 kHz), above which the performance converges and the unfilled inductor begins to perform better than the ferrofluid versions. At the low end of the range (100 kHz), the maximum power transfer efficiency was found to be 53.2 ± 0.7% for the unfilled inductor and 57.6 ± 0.9% and 57.3 ± 0.5% for the EFH1 and EMG900 ferrofluids respectively.

For comparison, our previous work [41] looking at stretchable wireless power systems with magnetic backplanes found that using a commercial ferrite magnetic backplane, as in the Wurth Electronics coil, increased maximum power transfer efficiency to a liquid metal coil by 8%, while adding a low permeability (relative permeability ˜3) ferroelastomer increased the maximum power transfer efficiency by slightly less (5%). The 4% improvement demonstrated with the ferrofluids here is therefore reasonable, and is significantly above the measurement uncertainty.

4. Conclusions

The ability to 3D print complex channels and fill them with magnetic and conductive materials is an important enabling technology, from customization to a specific application to making better use of the volume of compact electronics such as a cell phone. In this work, we presented a new method of incorporating magnetic and conductive materials into a 3D printed part, a multi-stage fabrication approach using the addition of both liquid metals and ferrofluids to create high quality factor components in an arbitrary 3D form factor. The resulting magnetic core inductors, transformers and a wireless power receiver coil incorporating ferrofluid are demonstrated to both increase the inductance density by nearly a factor of three over their unfilled counterparts, as well as to significantly improve power coupling in transformer and wireless power applications. As the first magnetic core 3D printed inductor in a room temperature process, this work is a meaningful step toward integration of power components into a variety of low temperature additive manufacturing processes. Future work will be focused on incorporating the demonstrated inductors into more complex power systems, as well as exploring the possibility of using the inductors for actuation of ferrofluids in 3D printed pumps and other fluidic systems.

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